TaskFlow

Plan • Organize • Achieve

Suggested Improvements

Personal & Collaborative Project Management Platform

PROGRAM	**DEPI**
DOCUMENT	**Suggested Improvements**
VERSION	**1.0**
PREPARED	**14 October 2026**
STATUS	**Proposed recommendations for team review**
AI	**Out of scope**

> **Note:** This document lists proposed improvement areas based on TaskFlow's planned scope. These are team recommendations, not lecturer feedback. Any lecturer-specific recommendations should be recorded only after actual feedback is received.

1. Purpose

This document identifies practical ways to improve TaskFlow's usability, reliability, security, and maintainability. The recommendations are designed for a platform that can serve individuals, freelancers, small teams, and companies without forcing every user into a company-style workflow.

Because the project has a limited DEPI delivery window, improvements are grouped by priority. The team should focus first on correctness, access control, timer reliability, and the main end-to-end user journeys; optional enhancements can be planned for later releases.

2. Improvement Priorities

- **P1 — Important for the initial release:** Improves core correctness, security, or successful demonstration.
- **P2 — Valuable enhancement:** Improves usability or completeness after the core workflow is stable.
- **P3 — Future enhancement:** Suitable for a later version if time and resources permit.

Effort estimates are qualitative and should be confirmed by the team before work is scheduled.

3. Recommended Improvements

ID	Improvement Area	Recommendation	Expected Benefit	Priority	Relative Effort	Suggested Owner
SI-01	**Personal-first onboarding**	Make a private workspace and personal project flow available immediately after registration. Offer team, company, and client features only when users need them.	Keeps TaskFlow approachable for individual users and freelancers while allowing later growth into team collaboration.	P1	Low–Medium	Mostafa
SI-02	**Clear project-progress calculation**	Document one clear progress formula for the first release, including how cancelled, archived, blocked, and incomplete tasks are counted. Show task counts alongside the percentage.	Prevents confusing or misleading progress figures for both users and clients.	P1	Medium	Ahd, with Ahmed and Alaa
SI-03	**Reliable timer recovery**	Persist timer state on the server, define pause/resume rules, prevent duplicate active timers, and provide a correction process for interrupted or forgotten sessions. Keep an audit history for edits.	Protects time records from accidental loss or double counting and makes reports more trustworthy.	P1	Medium	Ahd
SI-04	**Strong client-data isolation**	Test access restrictions for every client-facing resource, including project pages, tasks, attachments, deliverables, comments, time summaries, and direct URLs. Keep internal notes separate from shared content.	Reduces the risk of exposing another client's or the team's private information.	P1	Medium–High	Eman and Ahmed, with Mostafa
SI-05	**Consistent status and approval rules**	Define allowed task-state transitions and separate task completion from deliverable approval. Store the deliverable version, reviewer, decision, timestamp, and feedback for every review.	Makes workflow status unambiguous and creates a reliable history of client decisions.	P1	Medium	Alaa and Eman
SI-06	**Usable empty states and quick actions**	Provide clear first-use guidance when users have no projects or tasks, and include a quick way to create a task from My Work or the dashboard.	Reduces setup friction and supports quick capture for individual users.	P1	Low–Medium	Mostafa, with Alaa
SI-07	**Integration contracts and ownership**	Agree on shared entity names, DTOs, status values, service interfaces, and event contracts before module integration. Keep migrations coordinated and use small pull requests.	Reduces integration defects and conflicts among the five developers.	P1	Medium	Mostafa coordinates; all members contribute
SI-08	**Accessibility and responsive usability**	Check keyboard navigation, visible focus states, labels, error messages, contrast, responsive layouts, and readable tables/charts. Use WCAG 2.1 AA as a target where practical.	Makes the application easier to use across screen sizes and for people who use assistive technology.	P2	Medium	All frontend owners
SI-09	**Notification controls**	Let users configure relevant notification types and avoid duplicate or noisy alerts. Keep in-app notifications distinct from optional email notifications.	Reduces notification fatigue while preserving important reminders.	P2	Medium	Mostafa, with each module owner
SI-10	**Gantt and dependency safeguards**	Start with a clearly defined set of dependency types, validate dates, prevent circular dependencies, and indicate when a schedule calculation is not supported.	Avoids presenting misleading schedule or critical-path information.	P2	Medium–High	Ahd

ID	Improvement Area	Recommendation	Expected Benefit	Priority	Relative Effort	Suggested Owner
SI-11	**Search and filter consistency**	Standardize filtering and sorting conventions across project, task, time-entry, and deliverable lists. Ensure global search returns only records the current user may access.	Makes the product easier to learn and reduces inconsistent behaviour across modules.	P2	Medium	Mostafa, with module owners
SI-12	**Data export and recovery**	Provide appropriate exports for user/project records and document a database backup and restore procedure for the chosen hosting environment.	Helps users retain their work and supports recovery from accidental loss or deployment problems.	P2	Medium	Mostafa coordinates; Ahmed and module owners support
SI-13	**Performance measurement**	Measure representative operations such as opening a board, saving a task, loading the client room, and generating a time report. Record test conditions and fix the largest bottlenecks first.	Replaces unsupported performance claims with repeatable evidence and improves responsiveness.	P2	Medium	Ahd supports measurement; endpoint owners fix issues
SI-14	**User feedback loop**	Run short usability sessions with an individual user, a freelancer, a team member, and a client where possible. Record issues, prioritise them, and retest after fixes.	Helps validate that the platform suits different user types rather than only the development team.	P2	Low–Medium	Mostafa coordinates; all members participate
SI-15	**External integrations**	Consider calendar synchronization, GitHub/GitLab links, or email-to-task only after the core application is stable and a target-user need is confirmed.	Can reduce duplicate work for users who already rely on external tools.	P3	Medium–High	To be assigned after approval
SI-16	**Offline/PWA support**	Consider an installable Progressive Web App and read-only access to previously loaded data. If offline edits are ever added, define conflict handling before implementation.	Improves convenience for users who frequently work across devices or with intermittent connectivity.	P3	High	To be assigned after approval
SI-17	**Advanced administration and subscription features**	Consider subscription management, usage limits, SSO, and advanced audit controls only if TaskFlow is later positioned as a commercial SaaS product.	Supports a future business offering without overloading the academic prototype.	P3	High	To be assigned after approval

4. Priority Recommendations for the DEPI Version

Given the project's schedule, the team should prioritise the following improvements before adding further modules:

1. **Secure access boundaries:** Verify that each user and client can only access authorised projects and records.
2. **Reliable time tracking:** Test timer start, pause, resume, stop, refresh/reconnect, and duplicate-action cases.
3. **Clear project and task statuses:** Define status transitions, progress calculations, and the difference between task completion and client approval.
4. **End-to-end integration:** Ensure personal workspace → project → task → timer → deliverable review works across the assigned modules.
5. **Usable initial experience:** Make it easy for a person to start alone, with team and client collaboration available as optional features.
6. **Test and document the result:** Report actual measurements, known limitations, and deferred features honestly in the final submission.

5. Improvements That Should Not Be Added Automatically

The following should remain optional unless a clear need and sufficient time are confirmed:

- AI, machine learning, or LLM-based features. They are outside the agreed project scope.
- Full subscription billing, invoicing, accounting, or payroll.
- Native mobile apps.
- Complex enterprise portfolio planning or advanced resource optimization.
- Numerous external integrations that would consume time needed for the core workflow.
- Advanced Gantt critical-path calculations unless the team can test them reliably.

This scope discipline protects the quality of the implemented features and helps the team deliver a dependable demonstration rather than a large set of incomplete screens.

6. Measuring Whether Improvements Worked

Use clear acceptance checks instead of relying on subjective statements. Examples include:

- A new individual user can create a project and task without inviting other users.
- Timer sessions remain correct after a browser refresh and do not double-count intervals.
- A client cannot open another client's project or download an internal attachment by guessing a URL.
- A deliverable approval is linked to a specific version and records the reviewer and time of decision.
- A project-progress percentage matches the documented formula on a known test dataset.
- Representative users can complete core journeys with minimal assistance; record the sample size and observed issues.
- Performance measurements include the tested operation, environment, sample size, and actual result.

7. Improvement Tracking Register

Complete this table as recommendations are accepted and implemented.

ID	Decision (Accepted / Deferred / Rejected)	Assigned Owner	Target Date	Evidence / GitHub Issue	Status
SI-01	To be decided	Mostafa	To be assigned	To be added	Open
SI-02	To be decided	Ahd / Ahmed / Alaa	To be assigned	To be added	Open
SI-03	To be decided	Ahd	To be assigned	To be added	Open
SI-04	To be decided	Eman / Ahmed / Mostafa	To be assigned	To be added	Open
SI-05	To be decided	Alaa / Eman	To be assigned	To be added	Open
SI-06	To be decided	Mostafa / Alaa	To be assigned	To be added	Open
SI-07	To be decided	Mostafa + all members	To be assigned	To be added	Open

8. Conclusion

The most valuable improvements for TaskFlow are those that make the core experience trustworthy and understandable: a personal-first starting point, accurate time records, secure client access, clear task and review states, consistent cross-module integration, and evidence-based testing. More advanced integrations and commercial features can be introduced in later versions after the core workflows have been validated.

9. Approval

Reviewer	Name	Status / Date
Team Lead	Mostafa Mohamed Mahmoud Mohamed	Pending team review
Team Representative	To be confirmed	Pending
DEPI Instructor / Reviewer	To be confirmed	Pending